$$\frac{1}{\alpha}+\frac{1}{\beta}+\frac{1}{\gamma}+\frac{1}{\delta}=\frac{-a}{b}+\frac{a}{b-2}=\frac{5}{12}$$

If b = –4 $\Rightarrow \frac{-a}{-4}+\frac{a}{-6}=\frac{5}{12} \Rightarrow \frac{a}{4}-\frac{a}{6}=\frac{5}{12}$

$\Rightarrow \frac{3a-2a}{12}=\frac{5}{12} \quad \Rightarrow \frac{a}{12}=\frac{5}{12} \Rightarrow a = 5$

10. x + y + z = 15, xy + yz + zx = 72, then prove that $3 \le x \le 7$. (1997)

Sol:

$y(x + z) + zx = 72$

$\Rightarrow (15 - z - x)(x + z) + zx = 72$

$\Rightarrow 15(x + z) - (z + x)^2 + zx - 72 = 0$

$\Rightarrow (z + x)^2 - zx - 15(x + z) + 72 = 0$

$\Rightarrow z^2 + x^2 + zx - 15x - 15z + 72 = 0$

$\Rightarrow z^2 + z(x - 15) + (x^2 - 15x + 72) = 0$

$z \in R \Rightarrow b^2 - 4ac \ge 0 \Rightarrow (x - 15)^2 - 4(x^2 - 15x + 72) \ge 0$

$\Rightarrow x^2 + 225 - 30x - 4x^2 + 60x - 228 \ge 0 \Rightarrow -3x^2 + 30x - 63 \ge 0$

$\Rightarrow 3x^2 - 30x + 63 \le 0 \Rightarrow x^2 - 10x + 21 \le 0 \Rightarrow (x-3)(x-7) \le 0$

$\Rightarrow 3 \le x \le 7$

11. Give that 'x' is real then the solution set of the equation $\sqrt{x-1}+\sqrt{x+1}=1$

Sol:

$\sqrt{x-1}=1-\sqrt{x+1}$ (1988)

Squaring both sides and simplifying we get $1 = -2-\sqrt{x-1}$

again squaring and simplifying we get x = 5/4

but this 'x' does not satisify the original equation.

$\therefore$ No solution

(why it happens ? by squaring we get extraneous roots which may not satisify original equation.)

12. Let R be the set of real numbers.The number of functions $f : R \to R$ satisifying the relation$(x-1)^2 f(x) = 0$ is

1)Infinite 2)one 3)two 4)zero 5)none

Sol:

$f: R \to R$ satisfying the relation $(x - 1)^2 f(x) = 0 \to (1)$

BIT - TYPE QUESTIONS

1. If {a, b, c, d, e, f} = {2, 3, 4, 5, 6, 7} then the least possible value of ab + cd + ef is

   1) 50 2)52 3) 53 4) 60 5) 68

2. If $\sqrt{x+1} = 1 - x$ then which of thefollowing statement is correct?

   1) There are no solutions
   2) There are two solutions. The larger is greater than 2
   3) There are two solutions. The larger is less than or equal to 2
   4) There is only one solution. This solution is greater than 2
   5) There is only one solution. This solution is less than 2

3. The equation $\sqrt{x+7} + x = 13$ has

   1) no roots 2) one root 3) two roots 4) three roots 5) none

4. How many real solutions of the system $x - y = 3,\ x^2 - y = -1$ are there?

   1) none 2) 1 3) 2 4) 3 5) infinetly many

5. If $x^2 + 2x + n > 10$ for all real numbers x then which of the following conditions must be true?

   1) $n > 11$ 2) $n < 11$ 3) $n = 10$ 4) $n = \infty$ 5) $n > -11$

6. What is the largest integer k such that $\frac{3}{2} \cdot \frac{2}{1} \cdot \frac{1}{2} \cdot \frac{2}{3} \cdot \frac{3}{4} \cdot \ldots\ldots \frac{k}{k+1} \geq \frac{1}{8}$

   1) 20 2) 21 3) 23 4) 24 5) 26

7. What is the sum of all integers x that satisfy $-5 \leq \frac{x}{\pi} \leq 10$?

   1) 312 2) 324 3) 346 4) 376 5) 412

8. If $x > 5$, which of the following is smallest? $x \in Z^+$

   1) $\frac{5}{x}$ 2) $\frac{5}{x-1}$ 3) $\frac{x}{5}$ 4) $\frac{5}{x+1}$ 5) $\frac{x+1}{5}$

9. If *a, b, c, d* are four position numbers such that $\frac{a}{b} < \frac{c}{d}$, then which of the following is correct always

   1) $ab < dc$ 2) $a + c < b + d$ 3) $a + d < b + c$

7. If 'a' and 'b' are legs of triangle 'c' is hypotenuse then find $\log_a(c+b) + \log_a(c-b)$

EXERCISE -2

8. If log 3 = 0.477121, then find the value of $\log 3^{1000}$
9. If log2 = 0.3010 then number of digits in 2^{50}
10. log30 = 1.4771 then number of digits in 3^{100}
11. Prove that $\log_2 3$ is not a rational number
12. If $\log_{10} 15 = a$, $\log_{20}^{50} = b$ then show that $\log_9^{40} = \dfrac{5-b}{2ab+2a-4b+2}$
13. If $\log_{30}{}^{3} = a$, $\log_{30}{}^{5} = b$ then express $\log_{30}{}^{8}$ in terms of 'a' and 'b'
14. If $\log_{12} 27 = a$ then express $\log_6 16$ interms of 'a'
15. If $a = \log_6 15$ and $b = \log_{12} 24$ then express $\log_{125} 48$ in terms of 'a' and 'b'
16. If $\log_3 20 = a$, $\log_{10} 3 = b$ express $\log_5 30$ inters of 'a' and 'b'.
17. Calculate the value of $\dfrac{\log_2 24}{\log_{96} 2} - \dfrac{\log_2 192}{\log_{12} 2}$
18. If 'p' = -1.7321 , $q = \bar{1}.7321$ then what is the relation between 'p' and'q'. Ansq > p
19. Solve $\log_3 x + \log_3 y = \log_3 9 + \log_3 2$, $\log\log_{27}{}^{(x+y)} = 2/3$
20. Solve $\log_{16} x + \log_4 x + \log_2 x = 7$
21. Solve $\log_x 3 + \log_3 x = \log_{\sqrt{x}} 3 + \log_3 \sqrt{x} + 1/2$
22. Solve $\log_4(\log_2 x) + \log_2(\log_4 x) = 2$
23. Solve $\log_2 x + \log_2 y = 1$, $x + y = 3$
24. If $\log_{ab} a = 4$ then find the value of $\log_{ab}\left(\dfrac{\sqrt[3]{a}}{\sqrt{b}}\right)$
25. $\dfrac{\log_2 a}{4} = \dfrac{\log_2 b}{6} = \dfrac{\log_2 c}{3p}$ $and\ a^3b^2c = 1$ then find the value of 'p' .
26. If 'x', 'y' and 'z' are consecutive positive integers then show that 2logy = log (1 + xz)
27. If a, b, c are in G.P then show that $\log_a 10$, $\log_b 10$, $\log_c 10$ are in H.P

$\sec A + \tan A = a$ .....(1) , $\sec^2 A - \tan^2 A = 1$

$\sec^2 A - \tan^2 A = 1$, $\sec A - \tan A = \dfrac{1}{\sec A + \tan A} = \dfrac{1}{a}$ .....(2)

Add (1) and (2)

$2\sec A = a + \dfrac{1}{a} \Rightarrow 2\sec A = \dfrac{a^2+1}{a} \Rightarrow \sec A = \dfrac{a^2+1}{2a}$

In right angled triangleABC $AC^2 = AB^2 + BC^2 \Rightarrow (a^2+1)^2 = AB^2 + (2a)^2$

$AB^2 = (a^2+1)^2 - 4a^2 = a^4 + 2a^2 - 4a^2 + 1 = a^4 - 2a^2 + 1 = (a^2-1)^2 \Rightarrow AB = a^2 - 1$

$\therefore \sin A = \dfrac{a^2-1}{a^2+1}$

4. If 2sinA + 3SinB = 5 can 3sinA + 2cosB be evaluated ?.If so evaluate it ?(1994)

Sol :

2sinA + 3SinB = 5 , SinA ≤ 1 ,cosB $\le 1 \Rightarrow$ 2SinA ≤ 2 , 3cosB ≤ 3

If 2sinA < 2 and 3sinB < 3 then 2sinA + 3SinB< 5

$\therefore$ 2sinA + 3SinB = 5 is not possible for 2sinA < 2 and 3sinB < 3

2sinA + 3SinB = 5 is possible for 2SinA = 2 and 3cosB = 3

$\Rightarrow$ SinA = 1, cosB = 1 $\Rightarrow$ A = π/2 , B = 0

$\therefore$ 3sinA + 2cosB = 3(1) + 2(0) = 3

5. If for some angle *A* and for some real '*x*' it is given that $\sin A = x^2 - 2x + 2$ then find *x* and A. (1996)

Sol: $\sin A = x^2 - 2x + 2 \quad = x^2 - 2x + 1 + 1 = (x-1)^2 + 1$

If *x* > 1 or *x* < 1 then $(x-1)^2 + 1 > 1$

SinA > 1 which is impossible $\therefore x \ngtr 1$ and $x \nless 1$ $\Rightarrow$ SinA = 1 $\therefore$ A = $\pi/2$

$2\sec A = a + \dfrac{1}{a} \Rightarrow 2\sec A = \dfrac{a^2+1}{a}$

$(\text{Hyp})^3 > (\text{base})^3 + (\text{height})^3$

Continuing the process for ' n – 3' more times

$(\text{Hyp})^n > (\text{base})^n + (\text{height})^n$ for $n > 2$

7. In any triangle ABC **a** and **b** are the sides of triangle S =(½) abSinC then

1) $S \geq (a^2 + b^2)/4$ 2) $S \leq (a^2 + b^2)/4$

3) $S \geq (a^2 + b^2\text{-ab})/4$ 4) $S \geq (a^2 + b^2\text{-ab})/4$ (1995)

Sol :

$S = \frac{1}{2}\ \text{absin C} \leq \frac{1}{2}\ \text{ab}$ $(\because \sin C \leq 1)$

$A.P \geq G.P \Rightarrow (a^2 + b^2)/2 \geq ab$

$$\Rightarrow ab \leq \frac{a^2 + b^2}{2} \Rightarrow S \leq \frac{1}{2}\ ab \leq \frac{1}{2}\left(\frac{a^2 + b^2}{2}\right) \Rightarrow S \leq \frac{a^2 + b^2}{4}$$

8. Which one is greater between 31^{14} and 17^{18} (1995)

Sol:

$17^{18} = 17^{14}(17^4) > 17^{14}(16^4) = 17^{14}(2^{16}) = 17^{14}(2^{14})2^2 = (34)^{14}2^2 > 31^{14}$

Ans: 17^{18}

9. If p , q, r, s are positive real numbers then prove that

$(p^2 + p + 1)(q^2 + q + 1)(r^2 + r + 1)(s^2 + s + 1) \geq 81\ pqrs$ (1998)

Sol: $p \in R^+ \Rightarrow \frac{p^2 + p + 1}{3} \geq (p^2.p.1)^{1/3}\ \because AP \geq GP$

$p^2 + p + 1 \geq 3p$

Similarly $q^2 + q + 1 \geq 3q$, $r^2 + r + 1 \geq 3r$, $s^2 + s + 1 \geq 3s$

Multiplying above inequalities

$(p^2 + p + 1)(q^2 + q + 1)(r^2 + r + 1)(s^2 + s + 1) \geq 81\ pqrs$

10. If a, b, c are not equal and a + b + c> 0 determine the sign of $a^3 + b^3 + c^3 - 3abc$ (2000)

Sol:

$a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$

If b = d = 1 or -1 then $c = \frac{-6}{4d}$ can't be integer

$\therefore$ b = d = 1 or -1 may combine with c = 0

b = 1, d = 1, c = 0 $\Rightarrow$ ab – 2cd = 3 $\Rightarrow$ a = 3

b = -1, d = - 1, c = 0 $\Rightarrow$ -a = 3 $\Rightarrow$ a = -3

$\therefore$ all possible solutions are 1.a = 1 , b = 3 , c = 1 , d = 0

2.a = -1 , b = -3 , c = -1 , d = 0

3.a = 3 , b = 1 , c = 0 , d = 1

4.a = -3 , b = -1 , c = 0 , d = -1

31. For how many integers 'a' ($1 \le a \le 200$) is the number a^a is a square

Sol :

Case1: If 'a' is even a^a can be written as $a^a = (a^{a/2})^2$ which is perfect square.

If $1 \le a \le 200$ then possible 'a' values are 100

Case2 :

If 'a' is odd a^a can be written as $a^a = (a^{(a-1)/2})^2.a$ is perfect square if 'a' is perfect square. Perfect squares of odd numbers lying between '1' and '200' are $1^2, 3^2, 5^2, 7^2, 9^2, \ldots\ldots\ldots 13^2$ which are '7'.

$\therefore$ Number of squares lying between '1' and '200' in the form of a^a are 107

32. A sequence of positive integers is given by $a_1 = 1$ and $a_n = \text{g.c.d}(a_{n-1}, n) + 1$ for n >1 then calculate a_{2002}.

Sol:

$a_{2002} = \text{g.c.d}(a_{2001}, 2002) + 1$

$a_{2001} = \text{g.c.d}(a_{2000}, 2001) + 1 \Rightarrow a_{2000} = \text{g.c.d}(a_{1999}, 2000) + 1$

$a_{1999} = \text{g.c.d}(a_{1998}, 1999) + 1$

$\text{g.c.d}(a_{1998}, 1999) = 1$ or 1999 because '1999'is a prime

and by induction we can observe that $a_n < n$ for n > 2

$a_n < n \Rightarrow a_{1999} < 1999$

$\therefore \text{g.c.d}(a_{1998}, 1999) = 1 \Rightarrow a_{1999} = 1+1 = 2$

$\Rightarrow a_{2000} = \text{g.c.d}(a_{1999}, 2000) + 1 = 2+1 = 3$

$\Rightarrow a_{2001} = \text{g.c.d}(a_{2000}, 2001) + 1 = \text{g.c.d}(3, 2001) + 1 = 3 + 1 = 4$

$\Rightarrow a_{2002} = \text{g.c.d}(a_{2001}, 2002) + 1 = \text{g.c.d}(4, 2002) + 1 = 2 + 1 = 3$

Since the expression is of 4th degree, the remaining factor which should be symmetrical in a, b and c must be of the form k(a + b + c).

Thus $a(b-c)^3 + b(c-a)^3 + c(a-b)^3 = L(a+b+c)(a-b)(b-c)(c-a)$

In order to find the value of 'k'

Put a = 1, b = -1, c = 2, we have

$1(-3)^2 - (-1)^3 + 2(2)^3 = k(1-1+2)(2)(-3)(1)$

$-27 - 1 + 16 = k(-12)$

$\therefore k = 1$

Hence $a(b-c)^3 + b(c-a)^3 + c(a-b) = (a-b)(b-c)(c-a)(a+b+c)$.

7. Find the factors of $a^4(b^2-c^2) + b^4(c^2-a^2) + c^4(a^2-b^2)$.

Sol:

Given that $a^4(b^2-c^2) + b^4(c^2-a^2) + c^4(a^2-b^2)$.

By putting $b^2 = c^2$,

we find that the given expression vanishes, therefore (b^2-c^2) is a factor of the expression.

Similarly (c^2-a^2), (a^2-b^2) are also the factors of the expression.

Now the product factors of $(b^2-c^2)(c^2-a^2)(a^2-b^2)$ is of 6th degree and the given expression is of 6th degree.

So the only factor if possible may be some constant factor.

Let some constant factor be k.

Putting a = 1, b = 2, c = 3, we have

$(4-9) + 16(9-1+81)(1-4) = k(4-9)(9-1)(1-4)$

$-5 + 128 - 243 = k(-5)(8)(-3)$

$-120 = k \cdot 120 \Rightarrow k = -1$

Required factors are $-(b^2-c^2)$, (c^2-a^2), (a^2-b^2)

$\therefore a^4(b^2-c^2) + b^4(c^2-a^2) + c^4(a^2-b^2). = -(b-c)(c-a)(a-b)(b+c)(c+a)(a+b)$.

If a, b $\in$ Z, a, b $\neq$ 0

$\Rightarrow \sqrt{2}a, \sqrt{3}b$ are dissimilar surds, they never be equal.

$\Rightarrow$ a = 0, b = 0 $\quad \therefore$ a = b = c = 0

13. Solve $x^{\sqrt{x}} = \sqrt{x^x}$

Sol:

$x^{\sqrt{x}} = x^{x/2}$

If x < 0 $\sqrt{x}$ is not defined .

If x = 0 $x^{\sqrt{x}} = 0^0$ is meaning less .

$\therefore$ x = 0 is not in the domain .

If x = 1 is obvious root

If x $\neq$ 0, 1 then $x^{\sqrt{x}} = x^{x/2} \Rightarrow \sqrt{x} = \frac{x}{2}$

i.e. $2\sqrt{x} = x \Rightarrow \sqrt{x}\,(2-\sqrt{x}) = 0$

$x \neq 0 \Rightarrow 2 - \sqrt{x} = 0 \Rightarrow \sqrt{x} = 2 \Rightarrow x = 4$ .Hence x = {1, 4}

14. Solve $\sqrt{2x-6} + \sqrt{x+4} = 5$

Sol:

Squaring both sides

$$2x - 6 + x + 4 + 2\sqrt{(2x-6)(x+4)} = 25$$

$$\Rightarrow 2\sqrt{(2x-6)\,(x+4)} = 25 - 4 + 6 - 3x = 27 - 3x.$$

Again squaring both sides and simplifying.

We arrive at the equation $x^2 - 170x + 825 = 0 \Rightarrow x = 5, 165$

Direct substitution of these values in the original equation shows that x = 5 is root and x =165 is not.

why x = 165 is not a root of an equation?

Reason: (Extraneous root appeared due to squaring the equation)

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$\therefore$ Number of pairs = 4

7. If a679b is a five digit number that is divisible by 72 determine 'a' & 'b'.

Sol :

a679b is divisible by 72 $\Rightarrow$ a679b is divisible by 9 & 8

$\Rightarrow$ a number is divisible by '8' if last three digits is divisible by '8' i.e

79b is divisible by '8' $\Rightarrow$ b = 2.

a679b is divisible by '9' if sum of digits is divisible by '9'.

i.e for some 'M' $\frac{a+6+7+9+b}{9} = M$

a + b + 22 = 9M $\Rightarrow$ a + 2 + 22 = 9M $\Rightarrow$ a + 24 = 9M

for M = 1, 2 'a' becomes negative.

For M = 3 a = 27– 24 = 3

for M = 4 a = 36 –24 = 12

hence a = 3 , b = 2 .(because 'a' cannot be more than '9')

8. Find all primes of the form n^3 - 1 for *natural number 'n'*.

Sol :

Notice that $n^3 - 1 = (n-1)(n^2+n+1)$

and if $n \geq 3$, both of the factors on the right hand side are integer exceeding 1 If n < 3 then i.e for n = 1 , 2 n^3 -1 = 0 , 7 respectively.

$\therefore$ **Ans n = 2**

9. Find the units digit in the product of

1) 243x 512 x 709 2) $513^{103}.8^{59}.6^{73}$

Sol:

1.Units digit in the product 243x 512 x 709 is same as the unit digit in the product of numbers present in the units place

i.e unit digit in the product 3x 2 x9 = 54 is '4'

Ans .**4**

Sol.

Units digit in the product of $513^{103}.8^{59}.6^{73}$ is same as the unit digit in the product $3^{103}.8^{59}.6^{73}$

$3^{103} = 3^{100}3^3$

$3^{100} = (3^4)^{25}$ ends with 1 and $3^3 = 27$ ends with 7

$\Rightarrow$digit in the units place of $3^{100}3^3$ is '7'

$8^{59} = (8^4)^{14}8^3$,$(8^4)^{14}$ends with '6' and 8^3 ends with '2'

$\Rightarrow$digit in the units place of $(8^4)^{14}8^3$is '2'

6^{73} ends with '6'